Prince Agyei Tuffour

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Education

Oregon State University, Corvallis OR
Master of Science in Mathematics; Specialize in ***Facial Recognition & Optimization***
Kwame Nkrumah University of Science and Technology, Ghana
Bachelor of Science in Mathematics; ***GPA - 3.92/4.0***

Skills

Languages: **Python**, Javascript, C++, SQL, Java, Typescript, Go, Rust
Tools & Technologies: Git, Nextjs, Redis, AWS, Google Cloud, Playwright, Tailwind CSS, **Keras**, Pandas, **ScikitLearn**, Matplotlib, Deep Learning, **Linux/Unix**, PyTorch, **Tensorflow**, Selenium, Flask, Django, RESTful APIs, Express, PostgreSQL, MongoDB, **React**, SciPy, Streamlit, **YOLO**

Work Experience

Software QA Engineer | ***dynaConnections Corporations*** March 2025 - Present

- Elevated product quality from 73% to **95%** reliability score and reduced post-deployment bugs by 60% (from 150+ to **¡60 monthly incidents**) by implementing comprehensive test automation frameworks covering 85% of critical user journeys.
- Decreased average bug resolution time by **50%** (from 48 to 24 hours) by establishing systematic regression testing protocols and creating **500+** detailed test cases that caught **200+** edge cases pre-release.

Lead Software Developer/Executive Board Member | ***Akomapa Health*** March 2025 - Present

- Scaled platform to serve 500+ global users across 15 countries by architecting and launching an end-to-end health management system with **99.7%** uptime, directly impacting non-communicable disease awareness and treatment.
- Boosted user engagement by 75% and increased monthly active users from 800 to 1,400 by leading a team of 3 developers to deliver **12+ feature releases** and maintain platform performance for doctors and medical volunteers.

Machine Learning Engineer - Intern | ***Cita Marketplace.com*** Summer 2022 & 2023

- Contributed to **$250K+** in additional revenue and increased conversion rates by 15% by deploying an ML recommendation engine that personalized product suggestions for 50,000+ monthly active users
- Improved search relevance accuracy from 68% to **88%** and boosted customer retention by 18% by building predictive models analyzing **2M+** data points, while reducing ML feature deployment time from 6 weeks to 4 weeks

Graduate Research Assistant | ***Oregon State University*** September 2021 - December 2023

- Achieved **92%** prediction accuracy across **100,000+** data points and accelerated computational processing by 25% (reducing runtime from 8 hours to 6 hours) by engineering ML models using SVM, decision trees, and gradient descent optimization
- Streamlined research workflow processing **50GB+** datasets by building automated Python pipelines that cut data preprocessing time by **40%** and produced 1 published research paper through advanced PCA and t-SNE visualization techniques.

Projects

GradGPT.pro (Microservices | TypeScript | OpenAI API | AWS)

- Leading development of an AI-powered platform to democratize college admissions guidance, reducing consultation costs by **95%** (from **$5,000-$15,000** to **$19-$129**) and making premium guidance accessible to middle-income families across 4 countries with a comprehensive **400+** university database
- Building scalable **microservices architecture** using Node.js, Typescript, and React to support **5,000+** projected users, implementing intelligent AI model routing with **OpenAI GPT-4** integration to optimize costs while targeting **99.5%** uptime and sub-500ms response times
- Generated 12K+ personalized essay reviews and research proposal feedback, helping **89%** of users improve application quality

DocuMind (Python | PyTorch | FastAPI | AWS)

- Architected intelligent document processing pipeline using PyTorch and FastAPI with custom OCR and NLP models, achieving **96.8%** accuracy in structured data extraction

- Automated workflows that previously required 200+ manual hours monthly by processing 10K+ unstructured documents including PDFs, images, and scanned forms

CodeExplain (Python | TypeScript | Azure OpenAI)

- Built automated code documentation system using Azure OpenAI and AST parsing, generating comprehensive documentation for 5K+ lines of code across 15 repositories
- Created architectural diagram generation feature producing Mermaid and PlantUML diagrams from code structure analysis

GPU Memory Profiler for TensorFlow & PyTorch (Python | Deep Learning)

- Contributed to GPU memory profiling tool optimizing deep learning workflows by visualizing model memory usage for TensorFlow and PyTorch.
- Enhanced debugging, training efficiency, and memory usage visualization by identifying bottlenecks, aiding various ML practitioners in improving model performance.